Fixed-Step Black-Box Compressor Parameter Recovery via Shared-Potential Difference Networks and Geometry-Constrained Training

Haoyan Wang

***Abstract*—Training strategies for audio-based parameter recovery guide neural networks to predict control updates from paired target and current audio. However, the strategies that use smoothed directional losses can favor larger updates even when the predicted direction is correct, compromising fine adjustment of compressor settings. Herein, we demonstrate that a geometry-constrained training strategy enables precise parameter recovery in a black-box dynamic range compressor. Exact cosine supervision removes the incentive to inflate aligned updates, while coarse-to-fine training introduces local derivative constraints and increases the weight of compression-ratio errors in coarse pairs. A shared-potential difference network applies the same mapping to target and current audio and predicts parameter updates by subtracting the corresponding outputs. The architecture enforces zero updates for identical inputs and antisymmetry under input exchange. The trained model recovers compression threshold and compression ratio together with attack time and release time in two groups of eight tasks using four feedback steps per task. Mean absolute parameter errors decrease by factors of 18.39 and 10.95 relative to the original Sobolev-trained baseline under the same step budget, and 15 of 16 tasks attain a maximum normalized parameter error of 0.02 or less. The resulting framework establishes precise parameter adjustment through audio feedback and provides a basis for automated calibration of black-box audio effects.**

***Index Terms*—Audio-based parameter recovery, Dynamic range compression, Fixed-step feedback control, Geometry-constrained training, Shared-potential difference networks.**

I. Introduction

Loss functions for audio-based parameter recovery guided neural networks in estimating the control settings of audio effects from sound [1]. Direct regression used known settings as targets [2, 3], while differentiable signal processing allowed networks to learn from discrepancies between rendered and reference audio [4, 5]. Shared encoders also provided a way to learn features for estimating the controls of a dynamic range compressor [6, 7]. Compression threshold and compression ratio shaped the relationship between input level and gain reduction, whose evolution over time depended on attack time and release time [8]. The response to these controls depended on the input signal, so a close audio match did not establish that the underlying settings were close. A controller with a fixed feedback budget needed to turn the difference between the current output and the target into a usable change in settings at each step. Directional supervision therefore mattered because a correction had to point toward the target without exceeding the required change.

Smoothed directional losses were used in preliminary training to compare predicted updates with known differences between compressor settings. The smoothing term prevented division by zero when the predicted update vanished. The modified normalization also made the directional score depend on the length of the predicted vector. An update that already pointed in the correct direction could therefore incur a lower loss if its length increased. This incentive competed with parameter supervision once the predicted correction exceeded the target magnitude. The discrepancy mattered near the target, where the desired corrections were small, and the smoothing term contributed more relative to the update norm. This behavior motivated a direction term that remained constant when the magnitude of an aligned update changed.

Geometry-constrained training separated an update's direction from the size of the correction. An exact cosine loss compared the predicted and target parameter differences through their normalized inner product. A squared error term on standardized parameter differences supplied magnitude supervision. The local derivative penalty encouraged predicted updates to track small changes in normalized compressor settings near the training centers [9]. Coarse pairs exposed the model to larger discrepancies before fine pairs placed greater emphasis on small corrections. Extra weight on compression-ratio errors in coarse pairs emphasized the control that accounted for the remaining failures in the earlier grid evaluation. The joint objective was expected to retain effective corrections far from the target while improving fine adjustment near it.

Here, we developed a geometry-constrained training procedure for parameter recovery in a black-box dynamic range compressor. Audio rendered at known compressor settings supplied the paired inputs, and the difference between the normalized settings served as the training label. A shared-potential difference network used the same mapping for both inputs and subtracted the four outputs for the current audio from those for the target audio. The 8,000-update schedule began with coarse pairs and introduced fine pairs and local derivative supervision in the second stage. The compressor rendered audio after each applied correction, and that output provided feedback for subsequent updates. Two groups of eight tasks were used to compare the final normalized parameter errors of the trained network with those of the original model trained with Sobolev supervision, with four feedback steps allowed for each model. The resulting controller was expected to bring compressor settings closer to

Corresponding author: Haoyan Wang. Haoyan Wang is with the Naval Architecture and Ocean Engineering College, Dalian Maritime University, Dalian 116026, Liaoning, China P.R. Email: haoyanwang72@gmail.com.

their targets within a fixed number of steps and support automated calibration of black-box audio effects.

II. Methods

A. Parameter recovery task and audio representation

Parameter recovery targeted the single-band compressor in Xfer Records Serum 2, version 2.0.16, hosted in the Virtual Studio Technology 3 format through a modified Spotify Pedalboard 0.9.22 build. Compression threshold and compression ratio were adjusted together with attack time and release time. A fixed source preset and effect-chain configuration supplied the audio. Processor states were read back after loading, and the verified parameter values supplied the training labels and evaluation targets. The four controls were represented in normalized coordinates between zero and one. Compression threshold retained its native normalized value. Compression ratio was mapped by dividing its natural logarithm by the natural logarithm of 99. Each time parameter was expressed in milliseconds and normalized by taking the natural logarithm of its ratio to 0.1 ms, then dividing by the natural logarithm of 10,000. These transformations mapped compression ratios from 1 to 99 and time settings from 0.1 to 1,000 ms into the unit interval. The target update was the normalized target parameter vector minus the normalized current parameter vector.

Each recording contained three seconds of stereo audio sampled at 48 kHz. Musical Instrument Digital Interface messages triggered note number 48 at velocity 96 after 0.25 s and released the note after 2.25 s. Audio processing used a buffer size of 2,048 samples. A short-time Fourier transform used a 1,024-sample Hann window and a hop size of 512 samples for each channel [10]. Spectral magnitudes were compressed by taking the natural logarithm of one plus 100 times the magnitude, then reduced by adaptive average pooling to a two-channel representation with 128 frequency bins and 128 time bins. The recorded data bank contained 140 audio states, with 128 assigned to training and 12 to internal development. Feature means and standard deviations were computed from the original 96 training states and retained throughout the subsequent training stages. Standard deviations were bounded below by 0.001 before normalization. The coarse sampling pool contained 8,360 directed pairs and 112 self-pairs. The fine sampling pool contained 144 directed pairs and 18 self-pairs drawn from two local neighborhoods. Self-pairs supplied zero-update targets, while fine pairs had a maximum normalized parameter difference of 0.02.

B. Shared-potential difference network

A shared-potential difference network applied the same convolutional mapping to the target and current audio representations. The first convolution transformed two input channels into 16 channels using a 5 × 5 kernel with stride two and padding two. Two subsequent convolutions used 3 × 3 kernels with stride two and padding one, producing 32 output channels at each stage. Gaussian error linear unit activations followed every convolution [11]. Average pooling with a 4 × 4 kernel reduced the final convolutional representation to 512 flattened features. A fully connected layer mapped these features to 64 hidden units, followed by a Gaussian error linear unit activation and a four-output linear layer. Each output was multiplied by a fixed parameter-specific scale derived from the root mean square magnitude of the corresponding training labels. The output scale was bounded below by 0.000001. The network contained 47,796 trainable parameters, and the final linear layer was initialized with zero weights and biases.

The predicted update was obtained by subtracting the shared mapping output for the current audio from that for the target audio. Identical inputs therefore produced a zero raw update, and exchanging the inputs reversed its sign. The shared intermediate output also canceled when two updates were added through an intermediate state, giving path additivity before clipping. The current parameter vector was used to construct the subsequent compressor command.

C. Geometry-constrained training objective

The training objective combined standardized parameter-difference regression with directional supervision. The regression term was the mean squared difference between the predicted and target updates after division by the fixed parameter-specific output scales. The mean covered all four output components and every row in the batch, including self-pairs.

Directional supervision used one minus the exact cosine similarity between the predicted and target updates in normalized parameter coordinates. This term compared update orientation while retaining invariance to positive rescaling of a nonzero prediction. The implementation evaluated the angular term when the predicted update norm exceeded 0.00000001 and the target update norm exceeded 0.000000000001. Other rows contributed zero angular loss while remaining in the batch denominator. Parameter-difference regression supplied the training gradient when the predicted update was zero.

Local derivative supervision was introduced at two training centers. Paired perturbations of each compressor control supplied a feature-space tangent, calculated as the difference between the two feature representations divided by the actual normalized parameter span. Automatic differentiation propagated these tangents through the shared potential mapping to form a four-by-four local parameter-response matrix at each center. The derivative penalty was the mean squared difference between these matrices and the identity matrix, averaged over both centers and all matrix entries. The derivative computation retained the graph required to propagate this penalty into the network weights.

Additional compression-ratio supervision was applied to coarse pairs. The standardized squared compression-ratio error was summed over the coarse rows and divided by four times the full batch size. Adding this penalty duplicated the original contribution of coarse compression-ratio errors while preserving the batch denominator. Fine-pair errors retained their original weighting. The total objective assigned unit coefficients to the regression term, angular term, and

additional compression-ratio penalty, with a unit coefficient for the derivative penalty during the second training stage.

D. Coarse-to-fine optimization

Training used 8,000 updates with a batch size of 48 and a random seed of 42. The first 4,000 updates sampled 48 rows from the coarse pool. The remaining 4,000 updates sampled 24 coarse rows and 24 fine rows per batch, with local derivative supervision active throughout this second stage. Sampling within each pool was uniform with replacement.

The adaptive moment estimation optimizer with decoupled weight decay, AdamW, used a weight decay coefficient of 0.0001 [12]. The learning rate was 0.001 for the first 2,400 updates, 0.0003 for the next 800 updates and 0.0001 for the remaining 800 updates of the coarse stage. The optimizer was reinitialized at the start of the second stage. Its learning rate was then 0.0001 for 2,400 updates, 0.00003 for 800 updates, and 0.00001 for the final 800 updates. The global gradient norm was clipped to one before each optimizer step. Evaluation used the final checkpoint after 8,000 updates.

E. Fixed-step feedback recovery and evaluation

Closed-loop recovery used the target recording and the latest audio rendered from the current compressor state. Each predicted update component was clipped between −0.3 and 0.3, added to the actual current normalized parameter value, and clipped to the unit interval. The resulting command was converted back to the native compressor coordinates and loaded into the processor. Identical target and current waveforms triggered a zero update.

Each commanded state was rendered in two independent plugin instances to check repeatability. The first recording supplied feedback for the next step, and the corresponding read-back parameter values defined the new current state. The target parameters were used for scoring, while the feedback controller generated subsequent updates from the target and current recordings. Every task received four feedback steps, and the fourth-step state supplied the endpoint score.

Offline evaluation covered the coarse and fine pair sets and three development groups containing 84, 62, and 76 directed pairs. Closed-loop evaluation used two groups of eight target–starting-state pairs. The final network and the original Sobolev-trained baseline used the same tasks and four-step feedback budget. The baseline retained the shared-potential architecture and two-stage curriculum, with standardized parameter-difference regression and local derivative supervision.

Parameter mean absolute error was calculated by averaging the absolute normalized errors across the four controls, followed by averaging across tasks within each group. Per-parameter errors were reported alongside the maximum absolute error for each task. Success required every normalized parameter error to be no greater than 0.02, with 0.005 used as the stricter threshold. The error-reduction factor was calculated by dividing the group mean absolute error of the baseline by that of the final network.

III. Results and Discussion

A. Geometry-constrained training

Local derivative supervision aligned the learned parameter response with the identity mapping near the training centers. Paired target and current recordings were supervised by their actual differences in compression threshold and compression ratio together with attack time and release time, as shown by Fig. 1a. The joint objective assigned separate roles to update magnitude and direction, with local response consistency and additional compression-ratio weighting incorporated as shown by Fig. 1b. The 8,000-update curriculum used coarse pairs during the first 4,000 updates and mixed coarse and fine pairs during the remaining updates, as shown by Fig. 1c. The archived comparison between the coarse-to-fine model and its derivative-supervised successor quantified the effect of the local constraint at two training centers. As shown in Fig. 1d, the mean squared deviation from the identity decreased from 0.1253 to 0.00008059. This reduction shows that the supervised mapping learned a local response in which small parameter changes produced corrections close to the corresponding changes in normalized control coordinates.

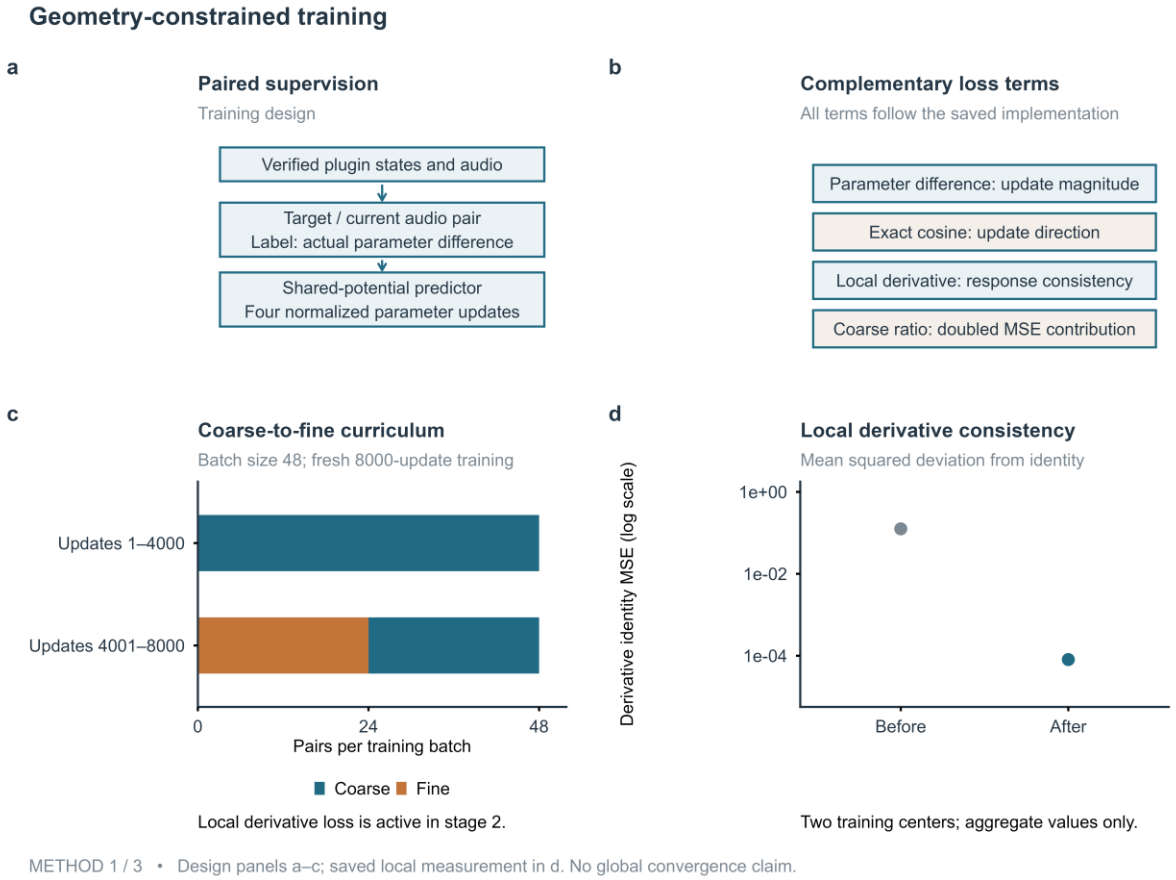


Fig. 1. Geometry-constrained training. (a) Paired audio and verified parameter differences. (b) Components of the training objective. (c) The 8,000-update coarse-to-fine schedule. (d) Mean squared deviation of the local response matrix from the identity at two training centers before and after Sobolev training.

B. Scale-invariant directional supervision

Exact cosine supervision removed the magnitude incentive embedded in the smoothed directional loss. The vector comparison separated errors in orientation from errors in update length, as shown by Fig. 2a. The smoothed loss remained positive for aligned vectors and decreased as the predicted norm increased, as shown by Fig. 2b. An aligned update with a norm of 0.01 incurred a smoothed loss of 0.2929 at a smoothing scale of 0.01, whereas the exact cosine loss was zero. The archived checks gave negative derivatives under radial scaling at all five tested norms for the smoothed expression and zero derivatives for the exact expression, as demonstrated in Fig. 2c. The negative derivative meant that increasing the length of an aligned prediction could lower the directional penalty. The corrected term therefore left magnitude calibration to parameter-difference regression while retaining supervision of update orientation.

The exact-angle candidate improved the learned corrections on the fine training pairs relative to the original Sobolev-trained baseline. The mean directional cosine increased from 0.96608 to 0.99990 across 144 directed pairs, as shown in Fig. 2d. The mean ratio between predicted and target update norms decreased from 1.01741 to 1.00249, approaching the ideal value of one. The corresponding mean absolute parameter error decreased from 0.00129993 to 0.00008230. The number of pairs with all four normalized errors no greater than 0.005 increased from 124 to 144. These results supported the exact directional objective as a complement to magnitude regression for learning corrections near the target.

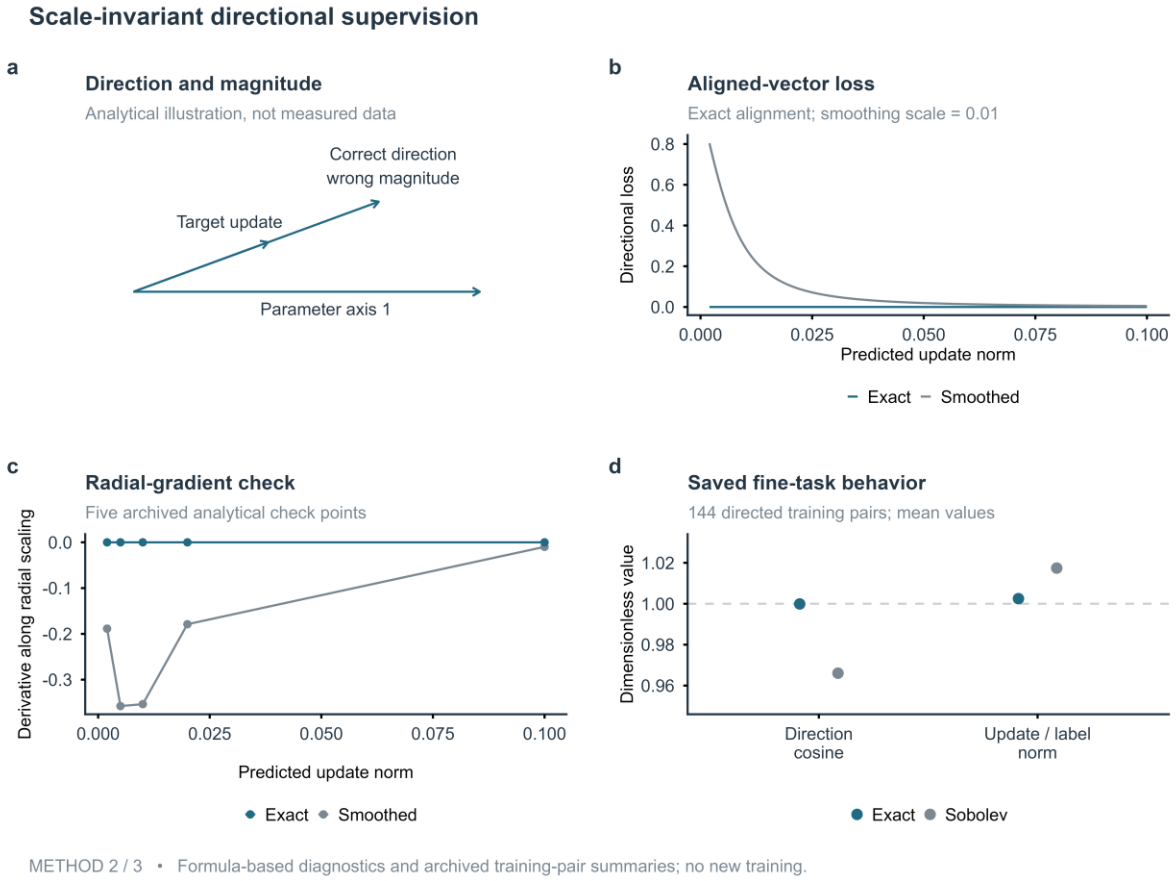


Fig. 2. Scale-invariant directional supervision. (a) Update direction and magnitude. (b) Smoothed and exact directional losses under exact alignment. (c) Radial gradients at five update norms. (d) Mean directional cosine and update-to-target norm ratio on 144 directed fine training pairs.

C. Compression-ratio weighting

Targeted compression-ratio weighting increased grid recovery success by addressing the dominant residual error. The exact-angle candidate left 44 failures among 76 grid tasks, with compression ratio exceeding the 0.02 threshold in every failure while the other controls satisfied the threshold, as shown by Fig. 3a. The additional penalty doubled the coarse-pair compression-ratio contribution and preserved the original fine-pair weighting and full-batch denominator, as shown by Fig. 3b. The number of tasks satisfying the all-parameter threshold increased from 32 to 46, as demonstrated by Fig. 3c. The mean compression-ratio error decreased from 0.02531 to 0.01849. Mean parameter error decreased from 0.01002 to 0.00803 in coarse pairs with unchanged compression ratio and from 0.01058 to 0.00853 in pairs requiring a compression-ratio change, as shown by Fig. 3d. The benefit therefore extended to settings where compression ratio was held constant as well as those requiring a change.

Compression-ratio weighting redistributed prediction accuracy across the recorded evaluation groups. Coarse training error decreased from 0.004809 to 0.004605, and grid error decreased from 0.009741 to 0.008685, as shown by Fig. 3e. The other two development groups increased from 0.01121 to 0.01182 and from 0.03736 to 0.04437. Fine-pair mean error increased from 0.00008230 to 0.00012089, while all 144 pairs retained the 0.005 success criterion, as shown by Fig. 3f. The final fine-pair error remained below the original Sobolev baseline value of 0.00129993. The added weight thus favored the dominant grid error while introducing an accuracy trade-off across the other groups.

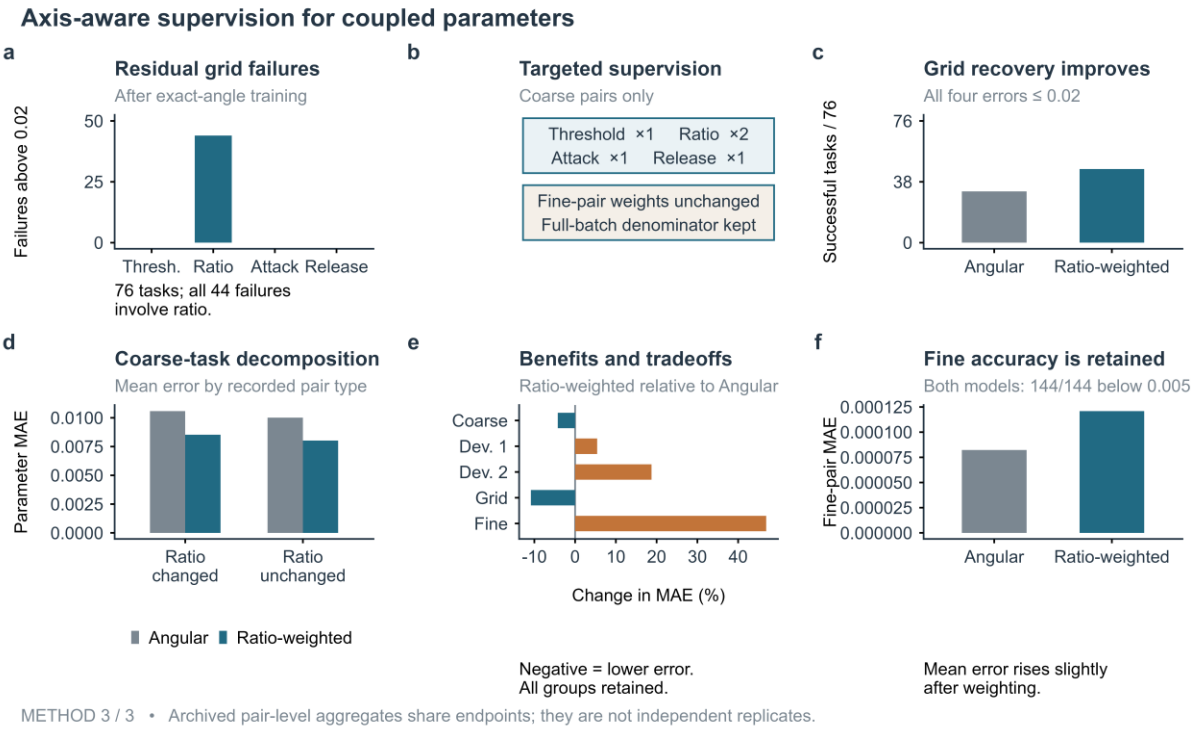


Fig. 3. Compression-ratio weighting. (a) Counts of grid failures by parameter after exact-angle training. (b) Additional weighting for coarse-pair compression-ratio errors. (c) Grid success before and after weighting. (d) Coarse-pair errors with fixed and varied ratio. (e) Relative changes in mean absolute error across five evaluation groups. (f) Fine-pair accuracy at the 0.005 error threshold.

D. Shared-potential difference architecture

Shared-potential prediction represented compressor adjustment as an explicit difference between two learned outputs. Target and current recordings passed through identical feature preparation, as shown by Fig. 4a. The common encoder and four-output head contained 47,796 trainable parameters, as shown by Fig. 4b. The current-audio potential was subtracted from the target-audio potential to obtain the update vector, as shown by Fig. 4c. Any constant offset shared by both potentials canceled in this subtraction. The predicted update was bounded and combined with the actual current settings to form the next parameter command, as shown by Fig. 4d. The network output therefore retained the interpretation of a correction in normalized control coordinates throughout the feedback loop.

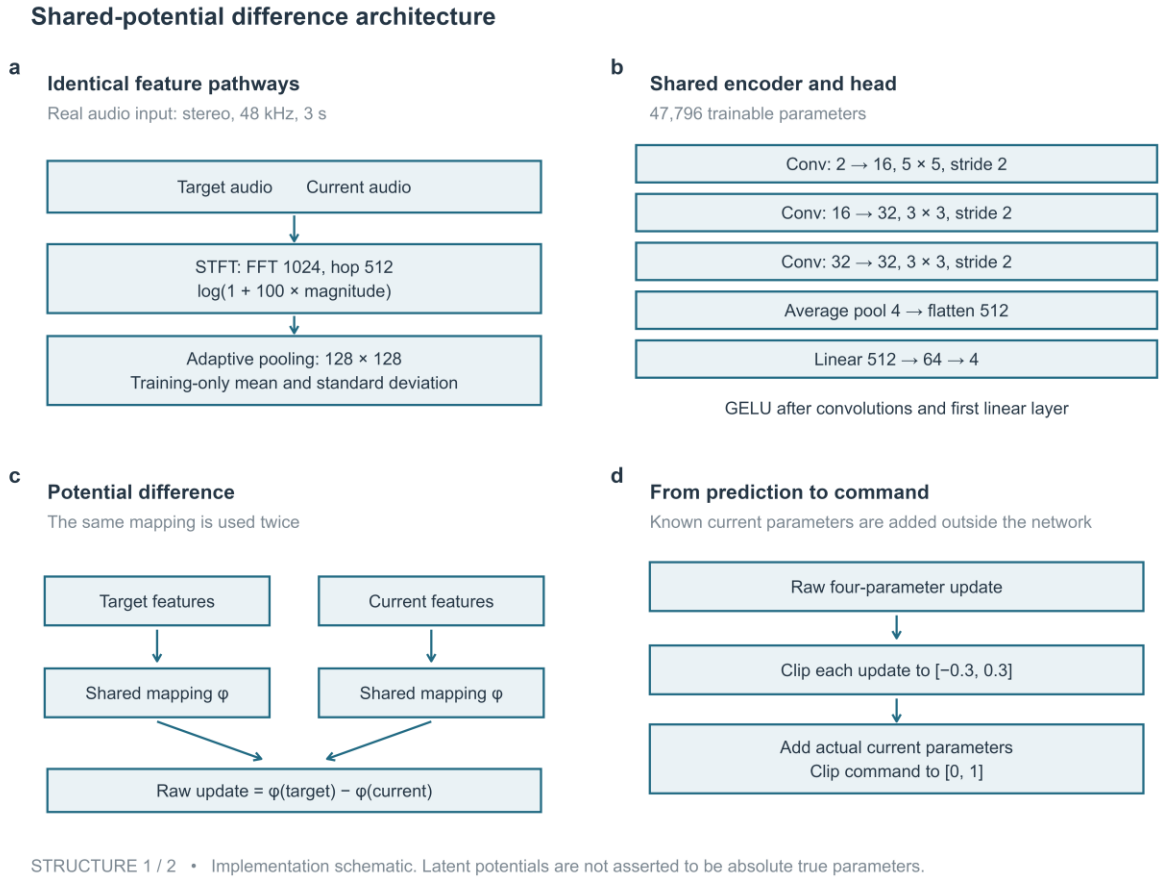


Fig. 4. Shared-potential difference architecture. (a) Identical preprocessing of target and current audio. (b) Shared encoder and four-output head. (c) Subtraction of the shared mapping outputs to predict the update. (d) Conversion of the predicted update into a bounded parameter command.

E. Structural consistency of parameter updates

The shared mapping imposed consistent relationships among raw parameter updates. Identical inputs produced a zero difference, as demonstrated by Fig. 5a. Exchanging target and current inputs reversed the sign of the update, as shown by Fig. 5b. Updates through an intermediate state added to the direct update because the intermediate potential canceled, as shown by Fig. 5c. These algebraic relations applied to raw predictions, while clipping could change executed-path additivity and floating-point calculations could introduce rounding residuals, as shown by Fig. 5d. Prediction accuracy depended on fitting the shared mapping to the parameter-difference labels and local response targets. Zero action and reversal consistency were embedded in the architecture, leaving the training objective to determine the mapping that supplied accurate corrections.

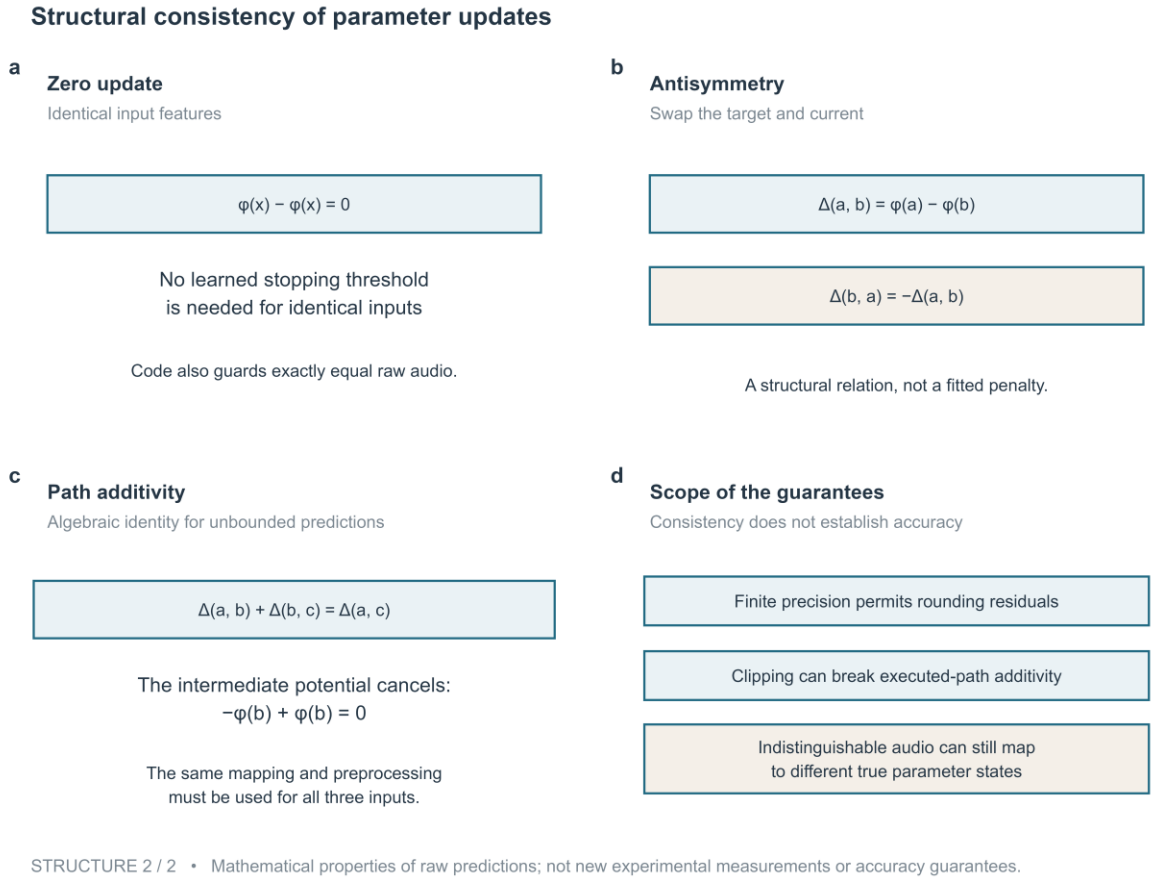


Fig. 5. Structural consistency of parameter updates. (a) Zero raw update for identical inputs. (b) Antisymmetry under input exchange. (c) Additivity through an intermediate state. (d) Effects of clipping and finite precision on these relations.

F. Fixed-step compressor parameter recovery

The final network achieved lower fixed-step parameter error than the original Sobolev-trained baseline in both feedback groups. Each network received the same four-step budget in two groups containing eight tasks each. Group 1 mean absolute parameter error decreased from an initial value of 0.06031 to 0.0006835 after four updates, as shown by Fig. 6a. Group 2 error decreased from 0.05759 to 0.0021466, as shown by Fig. 6b. The original baseline ended at 0.0125723 in Group 1 and 0.0234999 in Group 2. The ratios of baseline mean error to final-network mean error were 18.39 and 10.95 in the two groups, as demonstrated by Fig. 6c. The paired endpoints showed lower mean absolute parameter error for the final network in every task.

The lower endpoint errors increased the number of tasks meeting the prescribed recovery thresholds. The number of tasks with all four normalized errors no greater than 0.02 increased from five to eight in Group 1 and from four to seven in Group 2, as shown by Fig. 6d. The stricter 0.005 threshold was met by seven tasks in Group 1 and five in Group 2, compared with two tasks in each group for the baseline. The combined success counts were therefore 15 of 16 at 0.02 and 12 of 16 at 0.005. Final mean errors were lower for every parameter in both groups, as revealed by Fig. 6e. Compression-ratio mean error decreased from 0.02192 to 0.001528 in Group 1 and from 0.03082 to 0.004362 in Group 2. Compression ratio remained the largest residual component of the final network, consistent with its prominence in the offline grid failures.

Task-level endpoints also exposed the variation that remained after the four-step recovery sequence. The remaining failure at the 0.02 threshold occurred in Group 2 task 6, whose maximum normalized parameter error was 0.02700, as shown by Fig. 6f. Group 1 task 7 ended with a maximum error of 0.009194 and therefore satisfied the primary threshold while exceeding the stricter criterion. The saved stepwise counts for Group 2 reached eight successful tasks after step two, followed by six after step three and seven after step four. The endpoint comparison retained the fourth-step state for every task under the fixed evaluation budget. The reduction in mean error across all task endpoints demonstrated the performance of the controller combining the shared-potential network with geometry-constrained training. The resulting controller supported precise adjustment of compressor settings within a fixed sequence of audio-feedback updates.

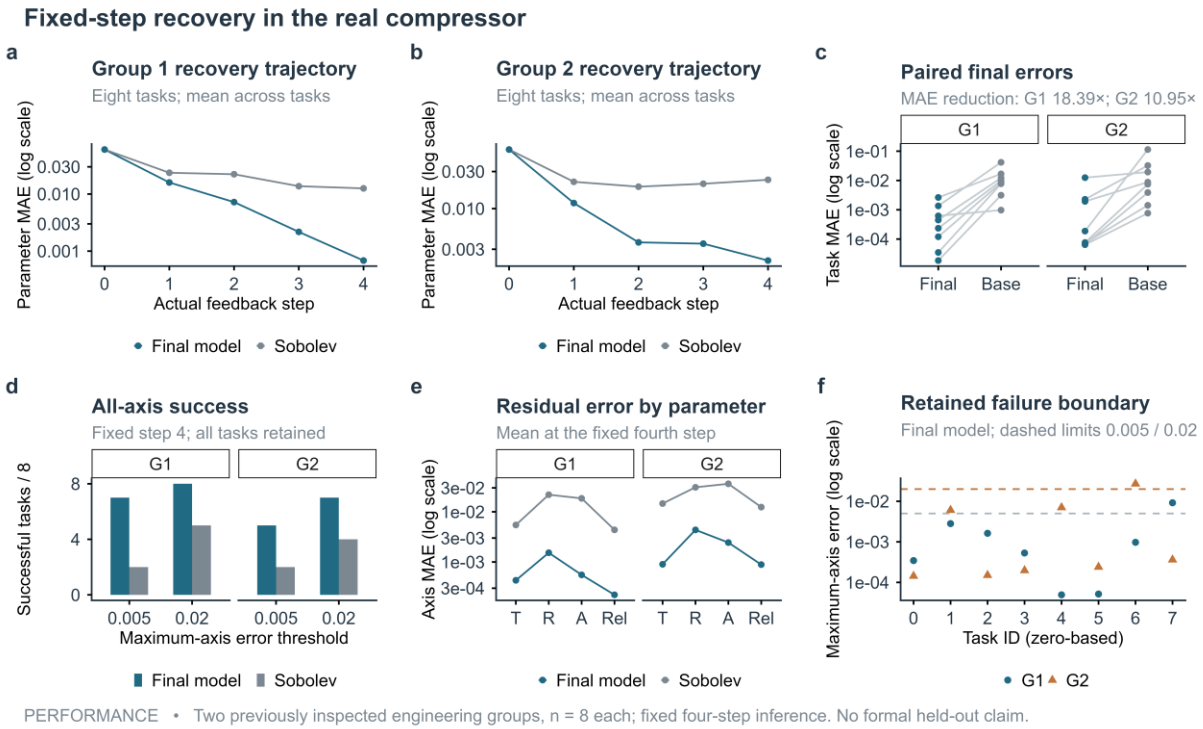


Fig. 6. Fixed-step compressor parameter recovery. (a, b) Mean parameter error over four feedback steps in two groups of eight tasks. (c) Paired endpoint errors for the final model and the original Sobolev-trained baseline. (d) Task success counts at normalized error thresholds of 0.02 and 0.005. (e) Endpoint mean absolute error for each compressor parameter. (f) Maximum normalized parameter error for each task with the final model.

IV. Conclusion

We demonstrate that geometry-constrained training enables precise recovery of black-box dynamic range compressor parameters from paired audio. The objective separates update direction from magnitude regression and introduces local derivative supervision during a coarse-to-fine curriculum, with additional weighting of compression-ratio errors in coarse pairs. Exact directional supervision improves fine corrections, while targeted compression-ratio weighting increases grid success from 32 of 76 to 46 of 76 tasks at the normalized error threshold of 0.02. The shared-potential difference architecture embeds zero raw updates for identical inputs and antisymmetry under input exchange through a common learned mapping. The final controller reduces mean absolute parameter error by factors of 18.39 and 10.95 relative to the original Sobolev-trained baseline in two groups of eight tasks under the same four-step feedback budget. All four normalized parameter errors reach 0.02 or less in 15 of 16 tasks and 0.005 or less in 12 of 16 tasks, demonstrating the precision achieved by the combined network and training design. Geometry-constrained learning therefore connects reference audio to effective parameter corrections, supporting automated calibration of black-box audio effects.

Acknowledgment

Existing experimental records supplied the reported numerical results.

Conflict of Interest

The author declares no conflicts of interest.

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